Indian Road Safety Campaign

Making Roads Safer!

Phase IV Final Report

Submitted by:

Aryan Varshney: 19CEB518

Siddharth Chauhan: 19CEB532

Kaushal Kumar Singh: 19CEB530

(Students of B. Tech 3rd in Civil Engineering, ZHCET, A.M.U.)

Under the Supervision of

Dr. M Rehan Sadique & Dr. Subhan Ahmad

(Assistant Professor, Deptt. Of Civil Engineering)

&

Ar. Amit Kumar

(Ph.D. from IIT Delhi)

Acknowledgment

The kind of contentment we realised; we could get after making the positive impact in the lives of people at our scale regarding road safety was something that made us pursue this project. Moreover, during the course, there were many technical as well as non-technical aspects that came into our knowledge after being a part of this opportunity. The overall journey has been fruitful and some of the following bodies/persons are worth appreciating, with the constant support of whom, the work till now could have been possible.

First of all, we would like to thank the Indian Road Safety Campaign (IRSC) which came up with the great initiative to bring commendable changes in the area of Road Safety in association with Ministry of Road Transport & Highways (MoRTH) and have been successful up to notable extent.

We would also like to express our sincere gratitude to Prof. Abdul Baqi (Chairperson, Department of Civil Engineering, ZHCET), Dr. M Rehan Sadique & Dr. Subhan Ahmad (Assistant Professor, Department of Civil Engineering) for their precious guidance throughout the project.

Last but not the least, the fellow teams are also thankful for joining hands for the projects as well as the visits to the Authorities for better & safer roads for the coming tomorrow.

Aryan Varshney

Siddharth Chauhan

Kaushal Kumar Singh

Certificate of Approval

Dated: 22/05/2022

This report is hereby approved as a recommendation for the Internship-cum-project work "IRSC Policy Internship to reduce road accidents" carried out and presented by Siddharth Chauhan: Team Leader (19CEB532), Aryan Varshney (19CEB518) and Kaushal Kumar Singh (19CEB530). As a manner to forward their suggestions regarding the project, I look forward for the positive changes to be made on the issues pertaining to Road Safety. However, the undersigned do not necessarily take responsibility for any conclusion drawn therein, but only approve the report to forward to the respective Authority.

Prof. Mohammad Altamush Siddiqui
Dean (ZHCET, AMU)

Prof. Abdul Baqi (Chairperson)
Department of Civil Engineering

Dr Subhan Ahmad
Assistant Professor
Department of Civil Engineering

Dr. M Rehan Sadique
Assistant Professor
Department of Civil Engineering

Index

1. Executive Summary

The upcoming lines indicate the possible Innovative as well as classic approach to some of the problems related to selected site: Ramghat Road (In Front of Pt. Deen Dayal Upadhyay Joint Hospital).

The project has been assigned by the **IRSC (Indian Road Safety Campaign)** working along with the **Ministry of Road Transport & Highways** (*MoRTH*) to minimize the Road Accidents as well as the possible factors to cause them. Overall guidance to the working team has been thoroughly provided by the faculty members, **Department of Civil Engineering, AMU**.

The suggestions to be presented to the Respective Authorities:

- The construction of trapezoidal humps at the marked location.
- Awareness Programs regarding Traffic Signs & Markings for the school students.
- Drawing of Traffic Markings and Installation of Traffic Signs (along with some given Innovative ideas regarding Tactical Urbanism).
- Enforcement regulation on the given spot.
- To cover or enclose the open sewer.
- To install Information boards (banners etc) at the given spot in order to aware the permanent stakeholders (particularly Shopkeepers & Vendors etc) regarding the knowledge of Traffic Signs and Markings along with their pictures.

Kindly refer the Index and upcoming slides for detailed explanation of these problems. Hopefully, the solutions are expected to yield some positive results regarding the concern of Road Safety…

2. Introduction

Aligarh, also known as the City of Locks, is one of the important cities in the Uttar Pradesh Western region. As UP has been amongst the Top 5 states in terms of Road Accidents in 2021 (a total of 37,537 road accidents in the state), it becomes a matter of concern for the respective authorities.

As data collected from S.P. Traffic Office, the number of Road Accidents in the District are as follows:

Year	Road Accidents	Death	Injured
2021	899	503	882
2020	881	506	807

With a population of over 41 Lakhs, the District Aligarh has a notable amount of Road users as well as recorded and unrecorded Road Accidents. With regard to our quest in minimizing the Road Accidents or to even reduce the chances for further causalities, we looked onto the various spots of the cities upon following the data from various Government offices and decided to work upon our current spot in front of Pt. Deen Dayal Upadhyay Joint Hospital, Ramghat Road, Aligarh.

The selected spot lies at a mere distance of 200 m from the Quarsi Chauraha and links up to the Dubey ka Parao Chauraha at a distance of 3.5km via Ramghat Road.

- Quarsi Chauraha: Amongst the busiest spots in Aligarh, the Crossroad extends/joins places like Dibai (at its far extension), Harduaganj and Etah Chungi Chauraha.

A clear depiction regarding the selected spot is further mentioned in the upcoming slides:

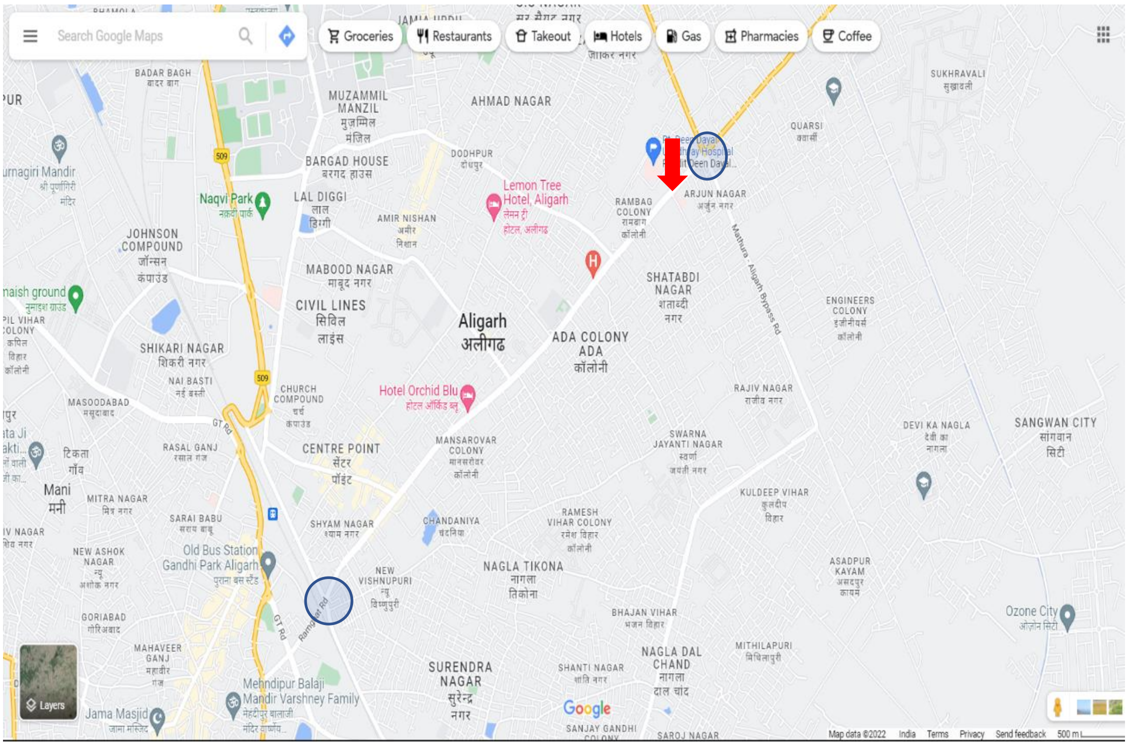


Fig1: Locality view of the spot (marked with the Red-Arrow). Image Source: Google Maps.

3. Bird Eye View of spot

In the above shown map, the two encircled points represent the Dubey ka Parao Crossroad and Quarsi crossroad while the downwards pointing arrow points towards the selected spot. A clear *Bird Eye view* is as shown in the following representation:

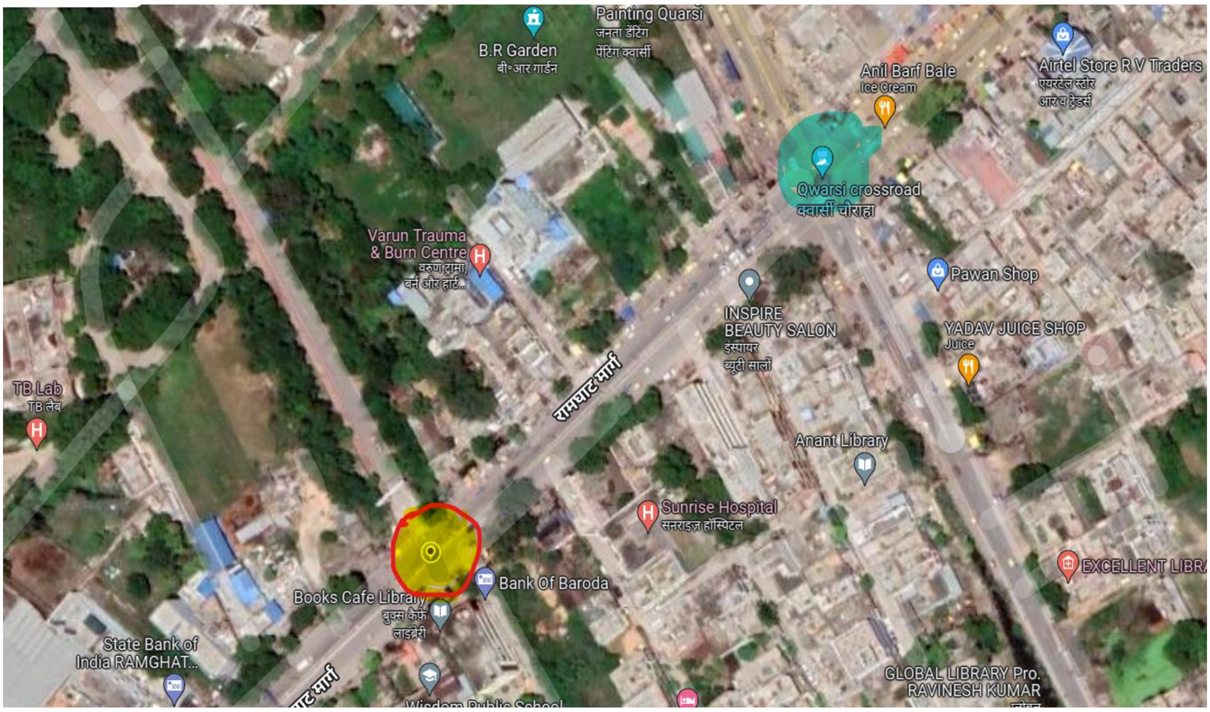


Fig2: Street view of the site marked in Yellow Colour spot (encircled by red). The cyan blue spot shows the Quarsi Chauraha. Image Source: Google Maps.

From the data collected through the reported cases registered in the FIR from the Quarsi Chauraha P.S. (in 2021)

- Total reported cases at the selected spot in 2021: 2
- Total reported cases at the Quarsi Chauraha: 3

Besides the reported data, there have been many unreported cases as well as minor collisions in day-to-day life at the spot, mainly observed due to the negligence of drivers.

Traffic Flow and Volume

Upon the survey of the spot on multiple days, we adopted the manual method of Counting along with videography of spot.

On general weekdays, the road was notably busy with Traffic Volume of **1250-1500 Passenger Car Unit (PCU**) in one hour (Keeping PCU & PCE in regard).

Weekends too didn't have much impact on the area and the traffic volume was more or less, the same.

The use of Intersection was also kept in mind, followed by observing the number of vehicles changing the lane as well as passing through the intersection and the value was calculated to be nearly 350 vehicles per hour for both Route 1 (extending towards Quarsi Chauraha) as well as Route 2 (vehicles approaching from Quarsi Chauraha towards the spot). The number also included the vehicles changing lanes from wrong orientation for which monitoring is required (mentioned in the upcoming slides)

4. Issues at the selected Spot

Fig3: Issues related to road safety at the selected site. Image Source: Google Maps & Snapshots

5. Some Snapshots of Site visits & Interviews

- Interview of Various Stakeholders: -

Fig4: Interview of School Watchman by Team Member Kaushal Kumar Singh.

Fig5: Interview of Traffic Cop by Team member Aryan Varshney at the Quarsi Chauraha towards regarding the spot stretch.
Image Source: Self Captured

Fig6: Interview of shopkeeper by Team Member Siddharth Chauhan.
Image Source: Self Captured

In all of these interviews, we tried to cover maximum aspects and survey questions along with their possible and feasible solutions to the same.

- Total Number of persons interviewed: 28

A brief idea of the survey during the process went through as follows:

- The profession of the people was written by the team member prior to their interview.
- Their frequency via spot.
- Whether evident of any accident or not.
- Their average speed of travel via this route.
- Issues pertaining to road safety on this spot.
- Knowledge of stakeholders regarding road Signs& Markings.
- Problems on road due to congestion of carriageway by
  - vendors,
  - Inappropriate parking,
  - Standing of autorickshaws at the intersection.
- Whether adequate lighting is present or not.
- Whether there is a need to provide Information (via banner etc) in order to get a better understanding of the road markings.
- Their agreement to the prepared solutions (after discussion with College Professors and in agreement with the mentor).

6. Other Visits:

Fig7: Visit to Traffic Police Lines, Aligarh

Fig8: With the S.H.O. of Jurisdiction

Image Sources: Self Captured

Fig9: Traffic volume during Weekday.

Image Source: Self-Captured

Manual Traffic Calculation were also being done at the same time.

7. Problems at the selected site

- ❖ **Overspeeding of Vehicles (mainly approaching from Quarsi Crossroad side).**
- ❖ **Lack of Enforcement & Monitoring.**
- ❖ **Lack of proper Road Markings & Traffic Signs.**
- ❖ **Problem of Road safety due to improper standing & halt by public (vendors, respective customers as well as Autorickshaws).**
- ❖ **Parking Issue.**
- ❖ **Open Sewer**

Fig 10,11,12 and13 respectively show the problems related to road safety at the selected spot.
Image Source: Self Captured

8. Proposed Solutions to the given problems:

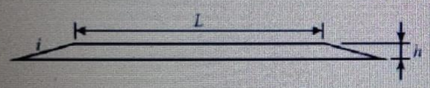
❖ **Installation of Traffic Calming device (Speed Breaker)**

As the approaching speed of most of the vehicles from the Chauraha towards the spot intersection is relatively higher than the speed limit, providing a speed breaker is the need of time to be implemented. The suggested type is the Trapezoidal Hump.

Trapezoidal Humps (IRC 99-2018:2.3.3.2):_For our selected location, this measure matches the criteria as:

- ✓ In the preventive measures, Speed Hump Type 2(flat top) is recommended for our Location in both:
  - Type of location: Having pedestrian crossing as well as Intersections.
  - Speed Limit of Road: Collector Road (speed Limit: 30kmph)
- ✓ Shall be cost effective as well.

  (Design guidelines mentioned in the IRC reference book).



Speed, km/h	Length L, m	Height h, m	Slope i
20	3.00–5.00	0.10*	1:6–1:10
30			1:10–1:15
50			1:20–1:30

Note: *the height of trapezoidal speed hump ranges from 0.08 m to 0.12 m depending on the situation

Fig14&15: Representation of Trapezoidal Humps. Image Source: IRC website&ext. link.

❖ **Creating a sense of awareness among & by School Students**

As the children & Students are the future leaders of the country, it shall therefore be helpful as well as advisable if the sense of awareness is brought right from their childhood itself. Under the guidance of the Traffic cops, the students of Wisdom Public School (Junior Wing) {located nearby to the spot} may visit the nearby location (i.e., either the selected spot or the Quarsi Chauraha) in a frequency of once a week a can be made aware of the Traffic Signs, markings as well as their importance. The home and extra co-curricular activities like programs etc may also be encouraged **which shall also plant a positive attitude regarding the Traffic Norms in the mind of their parents**. The idea, if successful, may also be implemented in various spots of the city along with active participation of other schools too.

❖ Enforcement Regulation & Penalized Region

As from the **IRC:SP:110-2017,** the ITS (Intelligent Transport System) is one of the proven ways to deal with the traffic in a smart manner, nowadays.

As already mentioned in the above problems,

- Overspeeding
- Unsafe Standing by vendors
- Parking on carriageway by Private vehicle owners
- Standing of Autorickshaw & E-Rickshaws at the spot

are some of those issues at the spot which needs to be taken in regard necessarily. As it is not feasible as well as practically possible to adopt the ITS at the selected spot, one of its *Subsystems*, i.e., Surveillance & Detection System is a need of time to be introduced at the spot to minimize the above-mentioned problems.

The Traffic police may also issue a challan for a suitable of fine for the violation.

- ***Installation of Automatic Number Plate Recognition Equipment:*** The ANPR is expected to strengthen the sense of discipline in the mind of citizens. The process includes to identify and communicate/circulate the data of over speeding to the respective office.

Fig16: Virtual Reprentation of ANPRs functioning. Image Source: Google

.

- ***Installation of Video Vehicle Detection (VVDs) or CCTVs:*** As the ANPRs seem not to be success in monitoring the roadside Vendors, the VVDs can be a good measure to monitor the congestion situation and deal with it accordingly.
  Fig: Automatic Number Plate Recognition Systems (Image Source: Google)

❖ Design of Road Markings and Installation of Traffic Signs

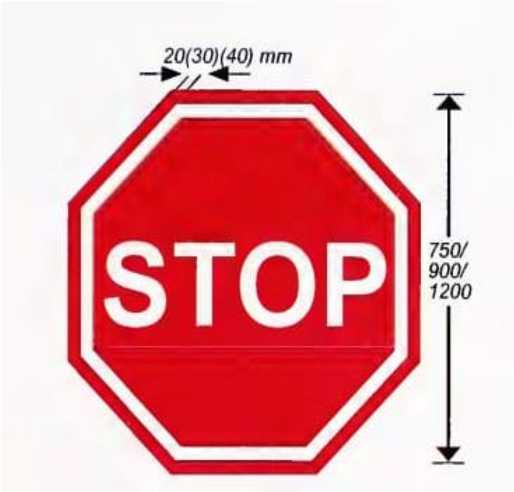
As the proper road markings and traffic signs are used to guide through the prescribed as well as informative road signals, these should be drawn and installed properly onto the road.

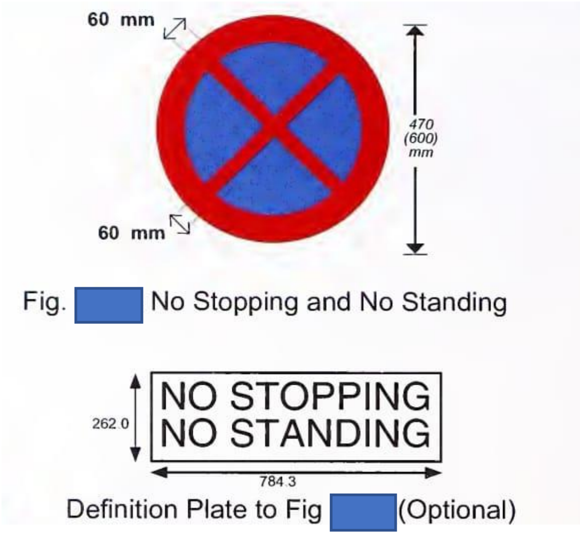
The drawing of Firm as well as dashed lines need not be drawn on the road as these are already present at the spot and its stretch.

Some of the recommended markings and Signs are as Indicated below:

- Traffic Signs: (As per IRC:67-2012)
  - Speed Limit: The speed limit sign board for the prescribed limit of 30km/h is to be installed on both of the ways of the roadside of the route. (Clause 14.9.8)
  - Stop Sign: Stop sign is recommended to use with the Stop Line Marking at the road coming from hospital. (Clause 14.5.5)
  - No Stopping and No Standing: As sudden halt of autorickshaws and e-rickshaws is one of the issues at the **Intersection**, it is therefore desirable to install a mentioned signboard to get rid of the problem. (Clause 14.8.4)

- ▪ <u>School Ahead</u>: As the selected spot also contains one of the famous schools of the city, the information is to be indicated by means of traffic signboard. (Clause 15.28)





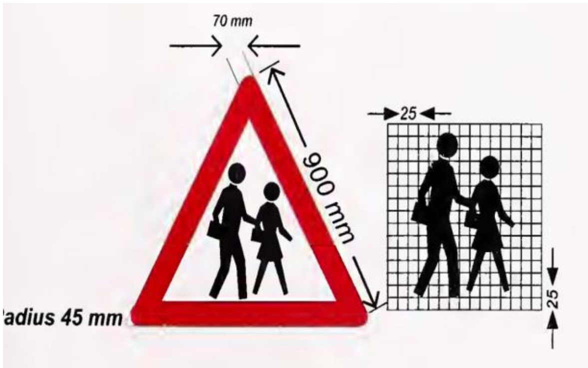


Fig17, 18, 19, 20: Represents Stop Sign. No Stopping & No Standing, Speed Limit Signboard and School Ahead Signboard respectively (Orientation in Clockwise direction).

- <u>Traffic Markings:</u>
    - ▪ <u>Speed Limit:</u> Speed Limit needs to be indicated in the form of traffic marking in order to indicate the prescribed value. (Clause 8.6)
    - ▪ <u>Pedestrian Crossing:</u> Pedestrian Crossings can be provided to facilitate the movement of notable quantity of pedestrians from hospitals as well as school nearby to the spot. The suggested ones are the 3D crossings shall be a better representation of visuals as well as are to be provided at the intersections. (Clause 11.3).

      Although a zebra crossing is already provided at the spot, but its positioning is not proper (maybe due to the extension of divider).
    - ▪ <u>Stop Line:</u> Single Stop Line can be provided before Pedestrian Crossing and at the road coming from hospital side with the above-mentioned Stop Sign Board. (Clause 6.1)

- Go Slow: The mentioned marking shall be provided to keep the vehicles slow as to indicate the presence of Speed breaker ahead. (Clause 8.6)
- School-Kept-Clear Marking: It should be provided in front of School to be kept clear of traffic. (Clause 13.2.1)
- Yellow Box Marking: It can be provided at the intersection with the No Stopping No Standing sign as mentioned above. (Clause 9.1.11)

Fig 21 & 22 represent the Speed Limit marking & "Slow" marking respectively.

Image Source: IRC Webpage

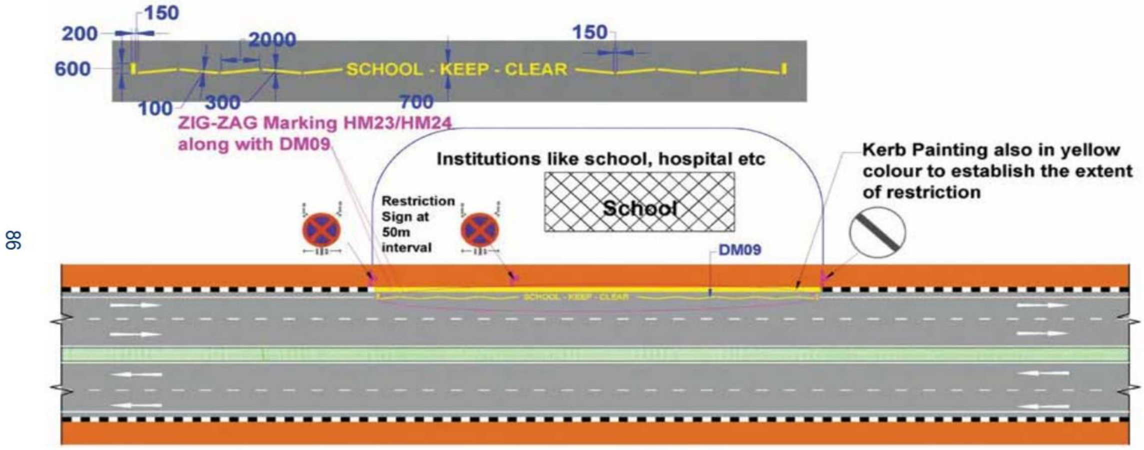


Fig 23 represents the Keep Clear Markings on the Road.

Fig24: position of Stop sign on the road.

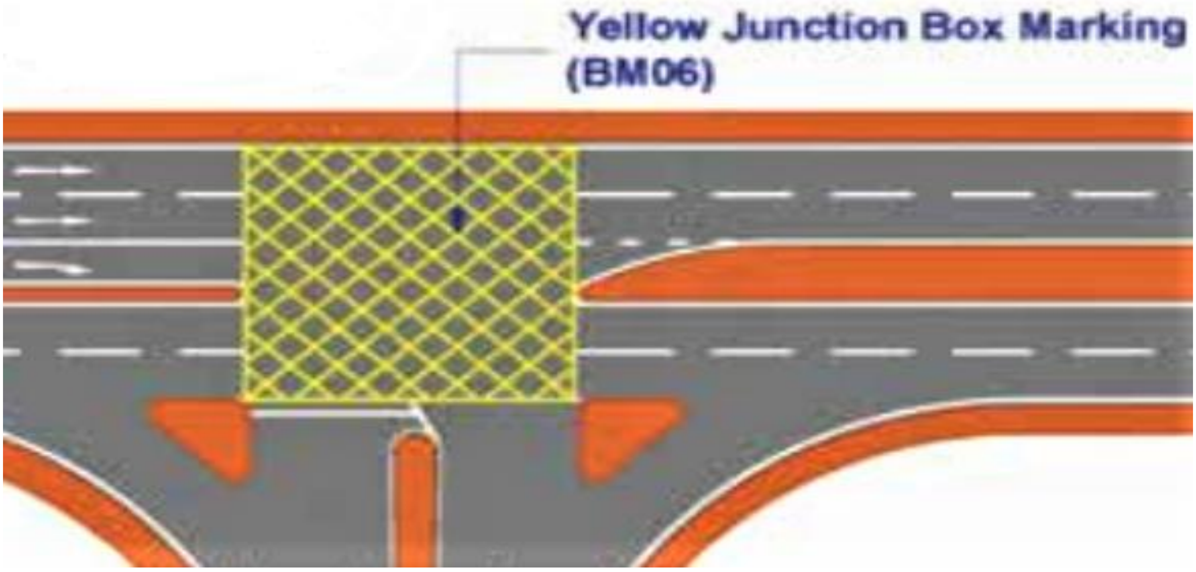


Fig25: Yellow Junction Box Marking

❖ An attractive alternative of stop line

As a part of tactical urbanism for efficient and economical solutions, drawings may impact the citizens simultaneously. In the proposed solution, the below shown figure can be an alternative of the Stop Line.

Fig26: An innovative alternative for stop marking

❖ Proper Enforcement of Yellow Junction Box & No Stop No Stand Sign

The solution is separately mentioned due to the current scenario of parking problem as well as halt of Commercial vehicles at the Intersection. The solution shall hopefully prove to be beneficial in controlling the congestion at the Intersection.

❖ Installation of Information boards regarding the Traffic Signs and Markings

Most of the relevant stakeholders (particularly vendors and shopkeepers) agreed upon the point to install an Information Board regardings the traffic signs and markings at the given spot.

The information may be same as that of mentioned in the previous slides along with their pictoral representation and the divider may be the suitable place to install them in the longitudinal direction of the road.

Some of the Position of Markings on the selected Spot

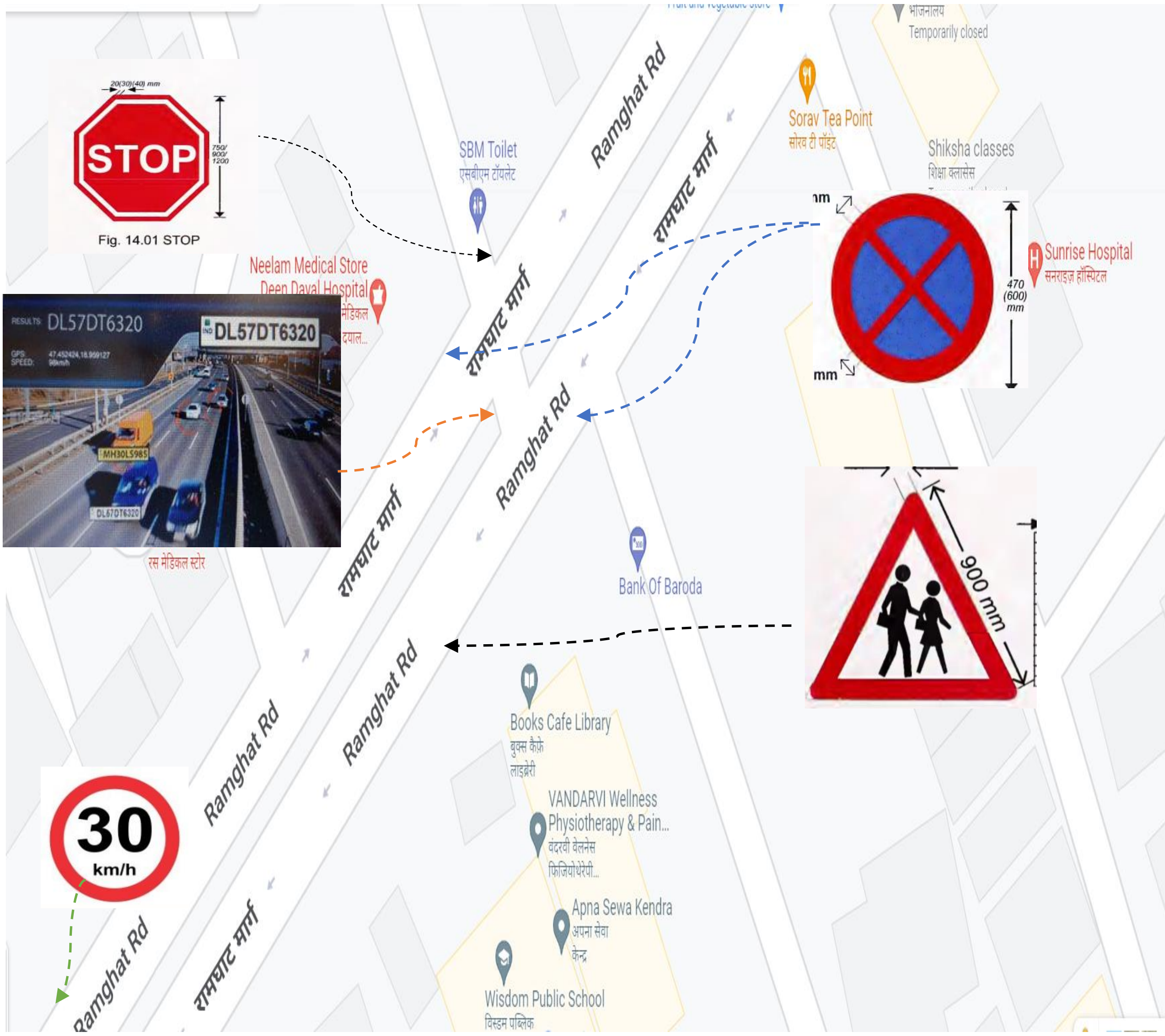


Fig27: Proposed Location of various traffic signboards on the given site. Image Source: Google Maps as well as other IRC webpage links.

Fig28: Proposed locations of Various Traffic Markings at the given site. Image Source: Google Maps as well as other IRC pics

❖ Covering the Sewer

As discussed mostly by the vendors, the open sewer near the spot (also indicated on **Page-6**) is one of the main problems at the area along with one of the minor unreported accidents caused by falling into it. The prescribed solution includes:

- Covering of small area of sewer by Prestressed Slab: The structure shall be strong enough to accommodate the parking of vendors at the place which will further result in the reduction of carriageway congestion at that particular point of the spot.
- Installing the Grating boundary at the perimeter of sewer: Being a more feasible solution between the two, this method can also be adopted, although, no additional benefit can be attained.

Fig29: Sewer to be covered
Image Source: Self Captured

❖ Foot Over Bridge

As from the recommendation, a small-scale foot over bridge may also be constructed at the crossing of the Major Street of the Site Intersection. The reason is the constraint to be not able to install proper Traffic Light due to the presence of system at merely 200m away. As the zebra crossing might not be the best solution, a foot over bridge may be constructed which shall also reduce the chances of Road Accidents.

Fig30: A Foot Over Bridge
Image Source: Google

“We hope that the recommended solutions shall be a step forward for the safety of road users. Suggestions are most welcome in this regard”

Thank You…